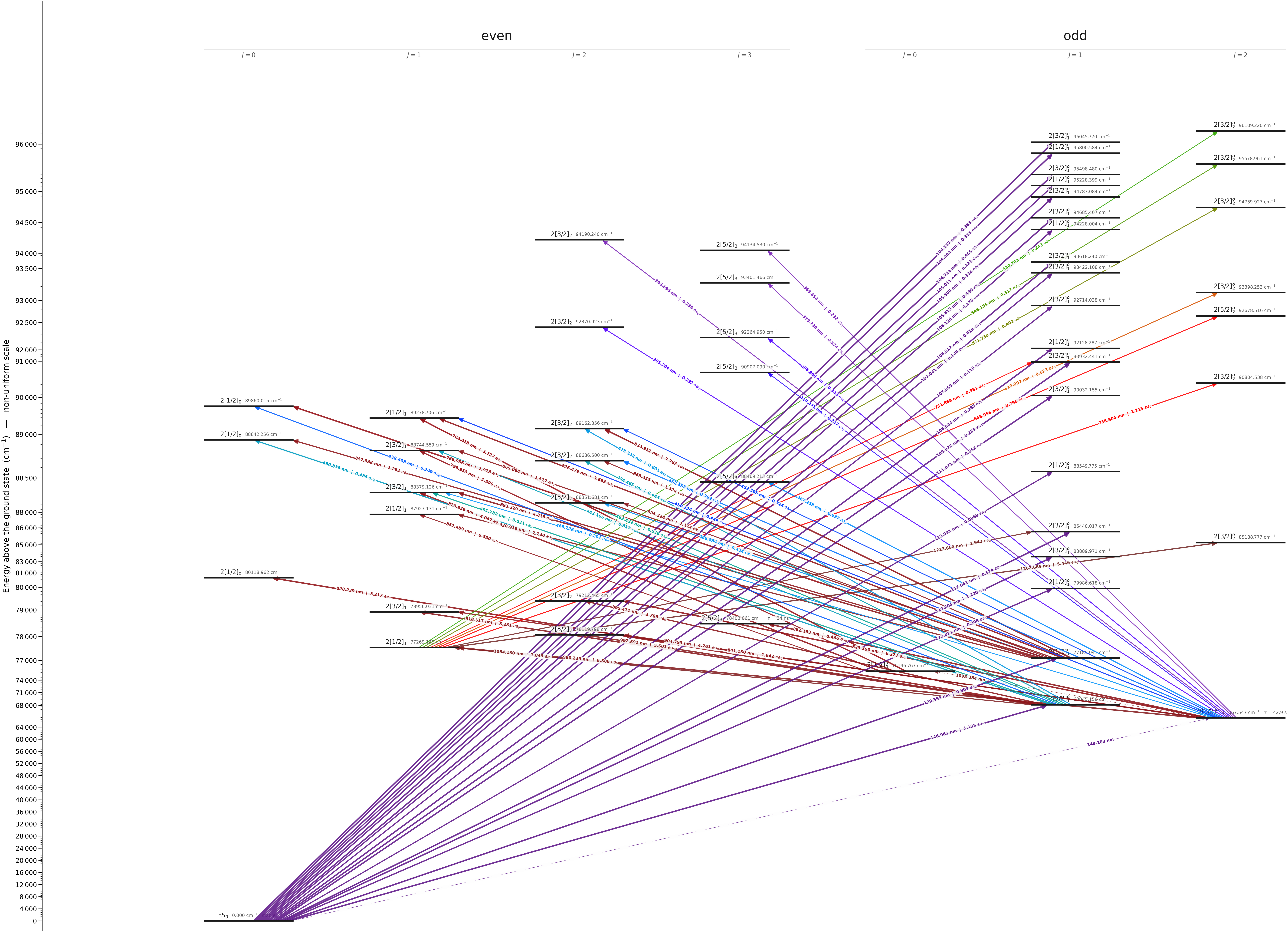
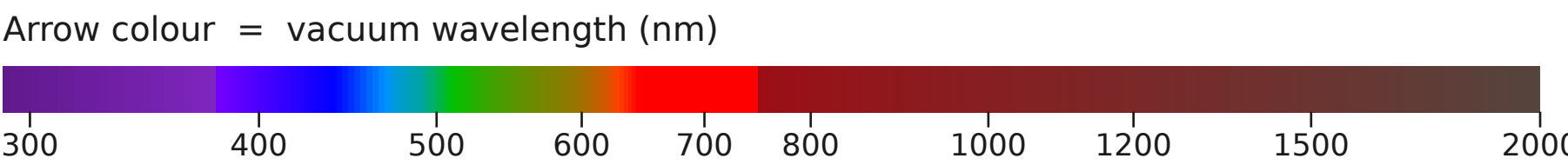


Xenon

Xe (Xe I). Strongest classified transitions below 2 μm (NIST): 51 levels, 72 lines.

Xe+ ionisation limit 97833.79 cm⁻¹ = 12.12984 eV (102.214 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|U||er||J\rangle|$ in $e a_0$
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	U er J⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
¹ S ₀ – 2[1/2] ₀ ^o	104.3835	–	2872.02930(10)	0.315	5.900e+07	–	–	104.4 nm VUV line: absolute calibration, hyperfine struct...
¹ S ₀ – 2[3/2] ₀ ^o	105.6128	–	2838.5989	0.580	1.930e+08	–	–	105.6 nm VUV line: absolute calibration, hyperfine struct...
¹ S ₀ – 2[1/2] ₁ ^o	106.1256	–	2824.8845	0.175	1.740e+07	–	–	106.1 nm VUV line: absolute calibration used for the ion...
¹ S ₀ – 2[3/2] ₁ ^o	106.8168	–	2806.6042	0.819	3.720e+08	–	–	106.8 nm VUV line
¹ S ₀ – 2[3/2] ₂ ^o	109.9718	–	2726.086012	0.283	4.080e+07	–	–	110 nm line measured by Ramsey-comb
¹ S ₀ – 2[3/2] ₂ ^o	146.9612	–	2039.9425	1.133	2.730e+08	–	–	147 nm resonance line: VUV-lamp first step of the optical...
¹ S ₀ – 2[3/2] ₂ ^o	149.1034	–	2010.6345	–	2.331e-02	3.71e-09	100	149.1 nm M2 decay of the metastable 3P2 state
2[1/2] ₀ ^o – 2[3/2] ₁	820.8591	820.6335	365.2179	4.047	2.000e+07	–	–	820.86 nm
2[3/2] ₀ ^o – 2[3/2] ₂	823.3897	823.1634	364.0955	6.277	2.860e+07	–	–	823.2 nm
2[3/2] ₀ ^o – 2[3/2] ₁	841.1501	840.9190	356.4078	1.642	3.060e+06	–	–	841.15 nm
2[3/2] ₀ ^o – 2[5/2] ₁	882.1832	881.9410	339.8302	8.436	3.000e+07	4.681	100	881.9 nm
2[3/2] ₀ ^o – 2[3/2] ₂	895.4709	895.2251	334.7875	3.789	8.100e+06	–	–	895.5 nm second step
2[3/2] ₁ ^o – 2[3/2] ₁	916.5168	916.2653	327.0998	5.231	2.400e+07	–	–	916 nm optical pumping into 6p[3/2]1, which decays to the...
2[3/2] ₂ ^o – 2[1/2] ₀ ^o	1095.3838	1095.0839	273.6871	–	–	1.243e-06	–	proposed optical frequency standard: two-photon clock tra...

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

¹²⁹Xe hyperfine structure (I = 1/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
2[3/2] ₀ ^o	-2384.4(12)	–	F=3/2: +3576.60 F=5/2: -2384.40
2[3/2] ₀ ^o	-959.1(7)	–	F=1/2: +959.10 F=3/2: -479.55
2[1/2] ₁ ^o	-5808.0(20)	–	F=1/2: +5808.00 F=3/2: -2904.00
2[3/2] ₁ ^o	-1308.9(6)	–	F=1/2: +1308.89 F=3/2: -654.45
2[3/2] ₂ ^o	-889.6(4)	–	F=3/2: +1334.40 F=5/2: -889.60
2[3/2] ₁	-4424.0(10)	–	F=1/2: +4424.00 F=3/2: -2212.00
2[3/2] ₂	-2740.4(9)	–	F=1/2: +2740.40 F=3/2: -1370.20
2[3/2] ₂	-2895(5)	–	F=3/2: +4341.90 F=5/2: -2894.60
2[1/2] ₁	1977.0(10)	–	F=1/2: -1977.00 F=3/2: +988.50
2[3/2] ₂ ^o	1657(12)	–	F=1/2: -1657.00 F=3/2: +828.50
2[1/2] ₂ ^o	-579(6)	–	F=1/2: +579.00 F=3/2: -289.50
2[3/2] ₂ ^o	913(23)	–	F=1/2: -913.00 F=3/2: +456.50
2[1/2] ₂ ^o	-4548(15)	–	F=1/2: +4548.00 F=3/2: -2274.00

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
¹ S ₀ – 2[1/2] ₀ ^o	104.383	136-134	213(12)
¹ S ₀ – 2[1/2] ₀ ^o	104.383	136-132	365(18)
¹ S ₀ – 2[1/2] ₀ ^o	104.383	136-131	492(20)
¹ S ₀ – 2[1/2] ₀ ^o	104.383	136-130	523(24)
¹ S ₀ – 2[1/2] ₀ ^o	104.383	136-129	678(24)
¹ S ₀ – 2[1/2] ₀ ^o	104.383	136-128	692(22)
¹ S ₀ – 2[1/2] ₀ ^o	104.383	136-126	869(25)
¹ S ₀ – 2[1/2] ₀ ^o	104.383	136-124	1069(22)
¹ S ₀ – 2[3/2] ₀ ^o	105.613	136-134	219(15)
¹ S ₀ – 2[3/2] ₀ ^o	105.613	136-132	354(20)
¹ S ₀ – 2[3/2] ₀ ^o	105.613	136-131	545(17)
¹ S ₀ – 2[3/2] ₀ ^o	105.613	136-130	529(30)
¹ S ₀ – 2[3/2] ₀ ^o	105.613	136-129	684(27)
¹ S ₀ – 2[3/2] ₀ ^o	105.613	136-128	726(20)
¹ S ₀ – 2[3/2] ₀ ^o	105.613	136-126	912(20)
¹ S ₀ – 2[3/2] ₀ ^o	105.613	136-124	1116(20)
¹ S ₀ – 2[1/2] ₁ ^o	106.126	136-134	207(12)
¹ S ₀ – 2[1/2] ₁ ^o	106.126	136-132	384(18)
¹ S ₀ – 2[1/2] ₁ ^o	106.126	136-131	531(23)
¹ S ₀ – 2[1/2] ₁ ^o	106.126	136-130	557(11)
¹ S ₀ – 2[1/2] ₁ ^o	106.126	136-129	675(14)
¹ S ₀ – 2[1/2] ₁ ^o	106.126	136-128	747(19)

Shift = ν(first isotope) – ν(second isotope).